

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Built a registry-governed semantic layer over BigQuery: YAML definitions plus SQLite vocabulary snapshots define legal dimensions, values, and table wiring — 16 governed demographic dimensions and 5 output types; a single LLM call maps a business question to strict JSON intent (the model never writes SQL), a deterministic gate validates and canonicalizes intent, and a deterministic compiler emits the SQL — same question, same answer, every time. New or changed definitions pass a human review gate before entering the registry.
- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with human-in-the-loop validation, 8 deterministic SQL rules, and a 23-case golden-query evaluation suite. Partnered with Product, Data Science, and business stakeholders to translate ambiguous business questions into agent-ready specs; approved by Sr. Director and product leadership and in active use by business teams. Eliminated the analyst as the bottleneck for routine asks.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.
- Replaced ~20 distinct recruitment workflows I had been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform. Stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates. A recurring 3-hour task now takes ~3 minutes of setup.
- Built and shipped to production an autonomous Jira ticket-resolution agent with retrieval over 11,000 historical SQL queries; closed 400+ data requests across 8 categories over 15 months, cutting routine turnaround from 30–60 minutes to under 10 minutes each. Stakeholders never knew an agent was answering them.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Data: BigQuery, dbt, SQL, Python, PySpark, Airflow, Databricks, MySQL

Viz / reporting: Tableau, Power BI (DAX, XMLA TMSL), LookML, Streamlit, Matplotlib, React Recharts

Engineering practices: CI/CD (pytest, version control), BigQuery scheduled queries, cron, JSONL state stores, FastAPI, Cloud Run, Slack webhooks

Semantic layer & evaluation: registry-governed semantic modeling, evaluation suites (golden sets, human-in-the-loop validation), decision logging / observability, context engineering

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL